

See also

[Travel Mode \[drop-down list\]](#) > [Move freely within area](#)

[Travel Mode \[drop-down list\]](#) > [Walk along footpaths](#)

[Travel Mode \[drop-down list\]](#) > [Beam to workplace](#)

Worker

Worker [object]

Use the object *Worker* for doing a job on a *Workplace* at a station.

Image



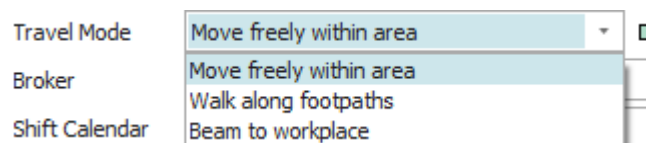
Description

The *Worker*-[WorkerPool](#)-[Workplace](#)-[FootPath](#)-concept refines the [Broker](#)-[Importer](#)-[Exporter](#)-concept.

As opposed to the *Exporter*, the *Worker* consumes **time** when **walking** to the *Workplace* at which he does his job. The *Worker* has the same statistics values as the *Exporter*, and in addition provides a statistics value for being **En-route to the Job**.

If your *Worker* does not walk, make sure that the importer on the tab [Importer](#) of the station is activated, on whose *Workplace* the *Worker* is to work.

In the [WorkerPool](#) you can select the [Travel Mode](#) with which the *Worker* gets to his work station:



- [Move freely within area](#)

The *Worker* walks freely within the area of the simulation model and walks around [obstacles](#), which you defined in 3D.

[Model a Worker Who Walks Freely Within the Model](#) gives a short introduction to the topic.

- [Walk along footpaths](#)

The *Worker* walks on the *Footpaths*, which you inserted between the *Stations/Workplaces*, to the station at which he is going to work.

- **Beam to workplace**

The *Worker* is beamed (teleported) to the *Workplace* to which he was brokered and at which he is going to work instantly, without using up any time at all, provided he cannot get to this *Workplace* on any *Footpath*. In this case the *Worker* functions as an **Exporter** with a capacity of 1.

You can use **Beam to workplace** to build a model which does not contain any *FootPaths*. The *Worker* then always jumps between the *WorkerPool* and the *Workplaces*.

You can also set the **Loading Time** and the **Unloading Time**, which the *Worker* needs for manually loading and unloading the station, in the **Workplace** attached to the station.

The *Workers* are created in the **WorkerPool** and they stay there if they do not work and are waiting for a work order. As a rule you type in **names** of the *Workers*, which you want to use in your simulation model, into the **Workers to Create** table. You can also drag a single *Worker* or any number of *Workers* from the *Class Library* onto the icon of the *WorkerPool* and drop them there. The *WorkerPool* appends a row to the **Workers to Create** table for this *Worker*. If it already contains a line for this *Worker*, the *WorkerPool* increments the amount by 1.

You can also create individual *Workers*, which you do not want to generate via the **Workers to Create** table, with the method **create** for a single simulation run. This only works in the *WorkerPool* though. Be aware that *Plant Simulation* deletes this *Worker* when you **reset** the model and then only creates the *Workers* from the **Workers to Create** table at the next **Init** event that takes place.

The **Broker** brokers the *Workers* to the individual work stations. As soon as a *Worker* can provide a service, the *Broker* sends him from his *WorkerPool* to the work station that ordered him. As soon as the *WorkerPool* receives a request for one of its *Workers*, which it can fulfill, the *Broker* sends the *Worker* to the *Workplace* which ordered him:

- If the *Worker* **walks freely** within the area of the simulation model, he walks on the shortest route to the *Workplace* and walks around obstacles you designated.

The *Worker* consumes time, when he walks freely within the area to the *Workplace*.

- If you modeled the work station using one of the material flow objects and the *Workplace* and connected the *WorkerPool* and the *Workplace* with a *FootPath*, *Plant Simulation* animates the *Worker* while he moves on the *FootPath* and while he works on the *Workplace*.

The *Worker* consumes time, when he walks on the *FootPath*.

- If you activated **Beam to workplace**, and you inserted a *Workplace*, and the *Worker*, who is supposed to work there, cannot reach it via a *FootPath*, *Plant Simulation* beams the *Worker* directly to the *Workplace* and animates him there.
- If you activated **Workers Can Work Remotely** and if you did not insert a *Workplace*, if no *Workplace* exists that supports this service, or if all *Workplaces* that support this service are occupied, *Plant Simulation* seemingly beams the *Worker* to the work station itself, where he will perform his task. It does not animate him there, but continues to animate him in the *WorkerPool*.

If a *Worker* is brokered to an *importer* for performing set-up, repairing or processing services and if this *Worker* then is assigned a failure, then the *importer* immediately places a new request with the *Broker*. Consequently, the respective operation can also be finished by another *Worker*.

If the *Worker* pauses or if a failure occurs when he finishes his work at the same time *Plant Simulation* does not delete the **DisruptionEnd**, **Out**, and **SetupEnd** events from the event list and does not request the *Worker* again after the failure or pause. This applies to set up, processing, and repair jobs.

Note

A paused *Worker* always returns to the *WorkerPool*. A failed *Worker* stays at the *Workplace*.

Note

A *Worker* who is automatically brokered does not return to the *WorkerPool* when he becomes paused or unplanned while he stays at the *Workplace*.

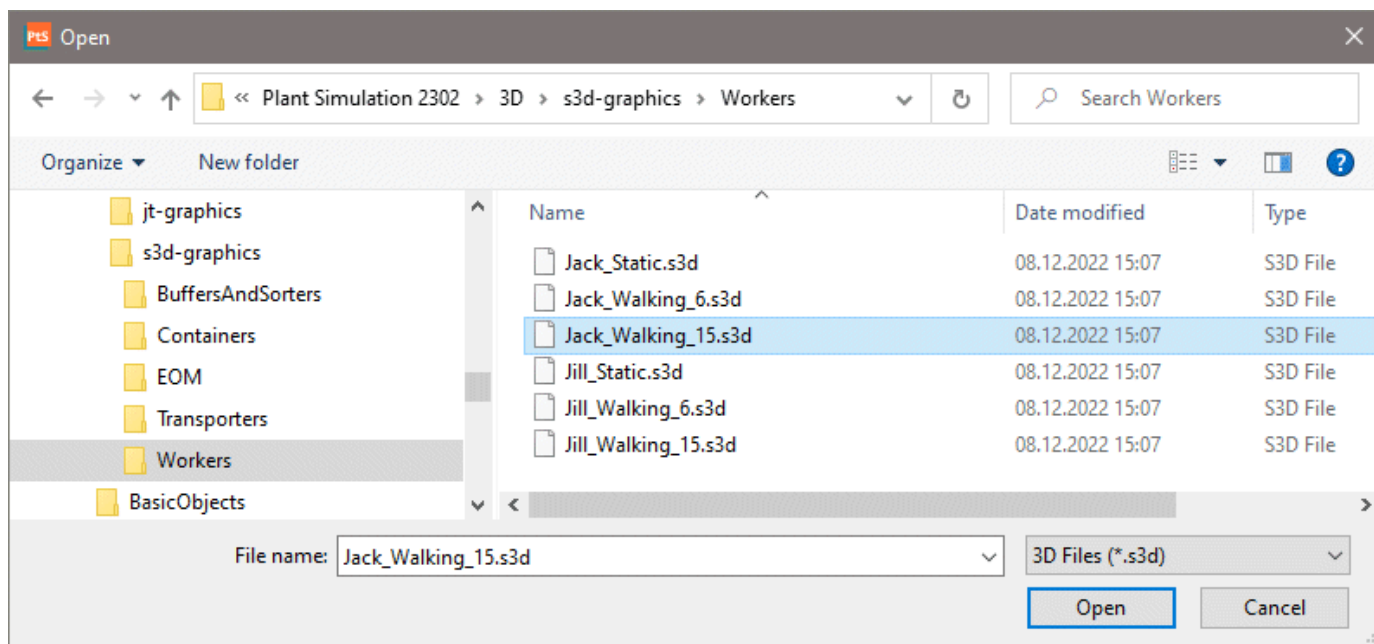
Note

A *Worker*, who is not automatically brokered stays, at the *Workplace* to which he was sent, no matter if you activated **Worker Stays Here After Completing the Job**/*WorkerStaysHere* or not in the *Workplace*.

Note

A *Worker* walks to the closest *Workplace* if several *Workplaces* are assigned to the work station.

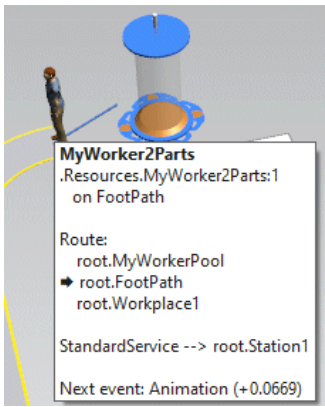
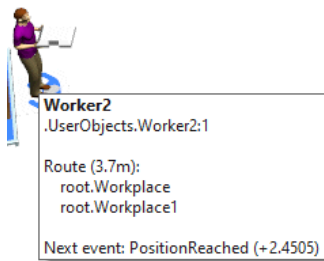
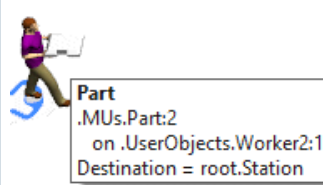
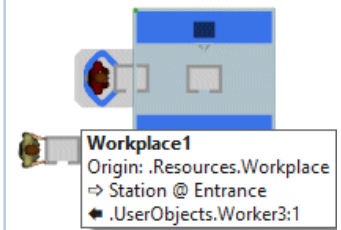
We offer alternative graphics for the *Worker*. Click **Exchange Graphics** in the 3D window of the *Worker* and select an alternative graphic.



On the **Tab MU Animation** you can set how the parts are distributed on the **Animation Area** of the *Worker*.

The **SankeyDiagram** visualizes the paths of the *Worker* who walk freely within the area.

To show a **tooltip** with information about the *Worker*, hover with the mouse over it.

			Workplace Assigned to Station
			Workplace Reserved for Worker (filled Arrow)
			Workplace Reserved for Worker on his Way (hollow Arrow)
Worker on Footpath with Route	Part Carried by Worker with Route	Part Carried by Worker Destination	
			

To show statistics of the *Workers* managed by the *WorkerPool*, drag it over a *Chart* and drop it there. Select the settings that you need.

Statistics Type ×

Group workers by:

Pool ▼

Occupancy with respect to:

Operational + failed ▼

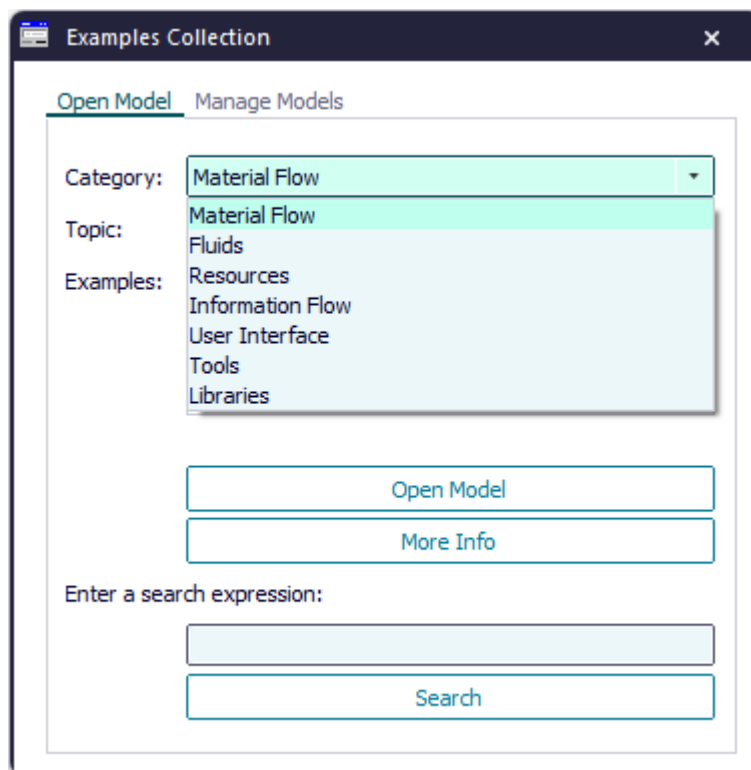
Operational + failed

Overall time

Add the Object to the Simulation Model

To add the object *Worker* to your simulation model, click  **Manage Class Library** > **Basic Objects** > **Resources** > **Worker** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page** > **Getting Started** > **Example Models** > **Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[How the Broker Mediates Jobs to the Worker](#)

[Model Workers and the Jobs They Do](#)

[Model a Worker Who Carries Parts Between Workplaces](#)

[Model Workers with Importer, Broker, and Exporter](#)

[How the Worker Finds the Shortest Route](#)

[How the Worker Decides Where to Work](#)

[How the Worker Carries Parts](#)

[How Workers Queue Up in Front of the Workplace](#)

[Show Worker Statistics in a Chart](#)

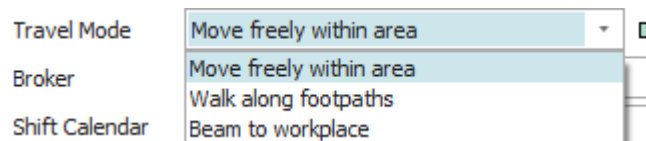
[Video on YouTube](#)

<https://youtu.be/HiPziA8kxc0?si=QnyvMaEQeVyaDoHo>

How the Worker Travels in the Plant

How the Worker Finds the Shortest Route

The **Travel Mode** determines how the *Worker* finds the shortest route from the *WorkerPool* and the *Workplaces* and between the *Workplaces*.



Remarks

You can select to make the *Worker*:

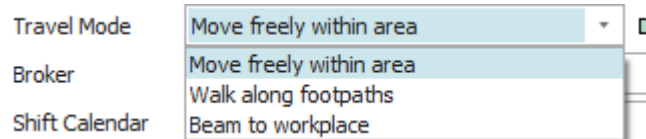
- [Move freely within area](#)
- [Walk along footpaths](#)
- [Beam to workplace](#)

See also

[How the Worker Decides Where to Work](#)

Move freely within area

Makes the *Worker* walk freely within the area of the simulation model and walk around the **obstacles**, which you defined.



Remarks

Plant Simulation automatically computes the shortest route between the *WorkPlaces* and between the *WorkerPool* and the *WorkPlaces*. It uses the **layout of the model** to compute the shortest route to a *WorkPlace*.

Note

We use the 3D layout to compute the route of the *Worker*. It normally represents the looks of your facility close to reality, meaning that the position, the dimensions, and the orientation of the machines and the fixtures of the building closely match the real plant.

To compute the shortest route, *Plant Simulation* uses information about those areas of the model, which the *Worker* cannot walk on or cross. These areas encompass all kinds of obstacles, for example walls, pillars, machines, conveyors, safety fences, etc. *Plant Simulation* automatically detects these areas. By default, we marked the *material flow objects*, the *animatable objects*, and the *graphics*, which belong to the *Frame*, as **Barred Areas**. *Information flow objects* and *user interface objects* by default are not **Barred Areas**. You can set if an object as an **obstacle** for the *Worker* under **Edit 3D Properties > Tab Graphics > Obstacle for the Worker**. For graphics, you can do this under **Edit 3D Properties > Tab Graphic Settings**, for example to allow the *Worker* to cross a *Track*. *MUs* do not provide this setting.



All settings, except for **None** automatically create a barred area.

To designate an area in the plant, which the *Worker* can enter, but should not, for example the pivoting area of a crane, you can manually create a **Barred Area**.

To allow the *Worker* to walk around or through obstacles, use a *FootPath*. Let's say that you want your *Worker* to **walk through a door** which in part of a wall that you defined as an obstacle. Then you can lay a *FootPath* through the door on which he can walk through it. The *FootPath* always has to be a little longer than the depth of the wall.

For the *FootPath* not the length, which you type in on the tab **Attributes** is decisive, but the **Length** of the *FootPath* in the scene. The distance covered of the *Worker* is determined by the starting point and the end point of the *FootPath* as well by the anchor points.

If the *FootPath* includes an **incline**, the *Worker* moves slower on the incline than on a flat surface. *Plant Simulation* achieves this by extending the imagined distance for the incline segments. When computing the length of an inclined segment, *Plant Simulation* triples the height difference.

Compare this example: A segment of a *FootPath* has a dimension of 1 m in the x-direction and 1 m in the z-direction, thus actually is 1.414 m long. The imagined dimension however is 1 m in the x-direction and 3 m in z-direction, so that *Plant Simulation* computes an effective length of 3.162 m.

The *Worker* always walks on the route, which has the shortest effective length and which he can thus cover in the shortest time. This can cause the *Worker* to take an actually longer but flatter route to the destination instead of a shorter route which contains inclines and gradients.

For computing the effective route length *Plant Simulation* triples the height difference. The resulting speeds of the *Worker* are realistic in general. The speed of the *Worker* decreases to a third, when the *Worker* moves vertically up or down, for example when climbing a **ladder**. The speed decreases to about half, when the *Worker* negotiates an incline of 38 degrees, for example a when walking on a **stairway**.

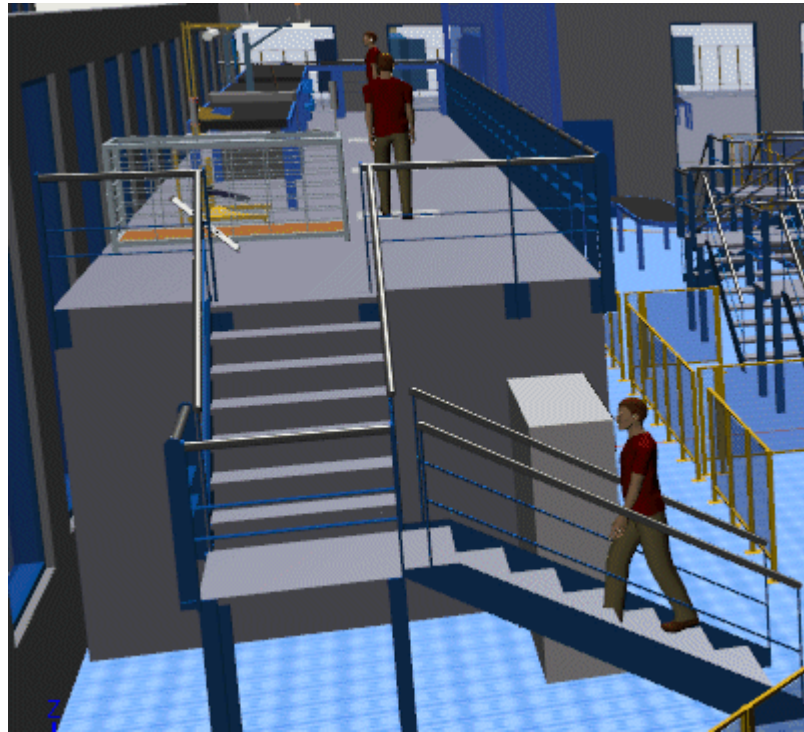
Note

The *Worker* walks slower on an incline than he walks on a flat surface.

Note

When the *Worker* **walks freely within the area**, he can only get to a different floor in the building by walking up or down a ramp on a *Footpath* or walking up and down **Stairs**.

The example model **Factory 51** demonstrates how you can use **Stairways**.



SimTalk

[WorkersTravelMode \[SimTalk\]](#)

See also

[Obstacle for the Worker \[drop-down list\]](#)

[Model a Worker Who Waits in Line](#)

[Model a Worker Who Walks Freely Within the Model](#)

[Set How Many Workers Are Created During Initialization](#)

[How the Worker Decides Where to Work](#)

[Set Home Location and Scope for the Worker](#)

[Choose the Nearest Worker \[check box\]](#)

[Edit 3D Properties > Graphics \[tab\] > Worker](#)

[Edit 3D Properties > Graphic Settings](#)

[Barred Area in 3D](#)

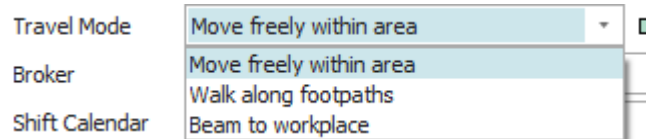
Move freely within area

Back to

Travel Mode [drop-down list]

Walk along footpaths

Makes the *Worker* walk on the *Footpaths*, which you inserted between the *Stations/Workplaces*, to the station at which he is going to work.



Remarks

To model and to simulate the distances between the *WorkerPool* and the work stations or between the work stations, insert *FootPaths*. *Plant Simulation* animates the *Workers* on the *FootPaths* when they walk toward the work stations or when they walk between the stations.

You can link several *FootPaths* with *Connectors* to create a network of *FootPaths*. The *Worker* walks within this network from his *WorkerPool* to the *Workplace*. Provided the *WorkerPool* and the *Workplace* are connected to the same network of *FootPaths*, the *Worker* walks from the *WorkerPool* to the *Workplace* and back on the shortest possible route.

A network of *FootPaths* consists of all *FootPaths* that are joined with *Connectors*, as well as the *WorkerPools* and the *Workplaces* that are joined with one of these *FootPaths* with a *Connector*.

The time it takes the *Worker* to walk from his *WorkerPool* to the *Workplace* depends on his **Speed** and the combined **Lengths** of the *FootPaths* he covers to reach his destination. *Plant Simulation* recognizes on which *FootPaths* the *Worker* walks if he exits a *FootPath* at another end than the one at which he entered it. If the *Worker* does not walk to his destination on a *FootPath* or if he is beamed to his destination, he does not consume any time at all.

SimTalk

[WorkersTravelMode \[SimTalk\]](#)

See also

[Set How Many Workers Are Created During Initialization](#)

[Edit 3D Properties > Tab Graphics > Obstacle for the Worker](#)

[Edit 3D Properties > Tab Graphic Settings](#)

Barred Area

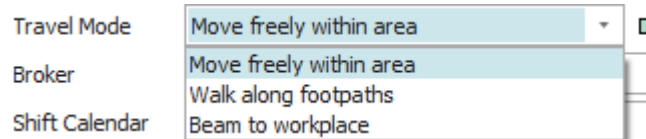
Walk along footpaths

Back to

Travel Mode [drop-down list]

Beam to workplace

Beams (teleports) the *Worker* to the *Workplace* to which he was brokered and at which he is going to work instantly, without using up any time at all, provided he cannot get to this *Workplace* on any *Footpath*.



Remarks

For the setting **Beam to workplace** the *Worker* functions as an *Exporter* with a capacity of 1.

You can use **Beam to workplace** to build a model which does not contain any *FootPaths*. The *Worker* then always jumps between the *WorkerPool* and the *Workplaces*.

SimTalk

[WorkersTravelMode \[SimTalk\]](#)

[teleportTo \[SimTalk\]](#)

[teleportToHome \[SimTalk\]](#)

[teleportToPool \[SimTalk\]](#)

See also

[Exporter \[object\]](#)

[Set How Many Workers Are Created During Initialization](#)

[Model a Worker Who Walks Freely Within the Model](#)

[Edit 3D Properties > Tab Graphics > Obstacle for the Worker](#)

[Edit 3D Properties > Tab Graphic Settings](#)

[Barred Area](#)

[Back to](#)

[Travel Mode \[drop-down list\]](#)

How the Worker Decides Where to Work

At times a *Worker* in the plant will have to decide if he continues working on a *Workplace* doing a job for which he has the correct skills or if he moves to another *Workplace* that requires his skills more urgently.

Remarks

While brokering, the *Broker* and the *Worker* proceed as follows:

1. The *Broker* first checks the **Priority** of the current **request** in the dialog of the *Worker*.
If the job has the highest **priority**, he stays on the current *Workplace* and finishes the current job.

The screenshot shows the 'Attributes' tab of the '.Resources.Worker:1' dialog. It contains the following fields and controls:

- Name:** Worker
- Label:** (empty)
- Number:** 1
- Stopped:** ☐
- Failed:** ☐
- Planned:** ☒ (selected)
- Priority:** 0
- Efficiency:** 100
- Speed:** 1.5
- X-Dimension:** 1
- Y-Dimension:** 1
- Z-Dimension:** 1
- Shift:** (empty)
- Services:** (dropdown menu)
- Service:** Processing
- Broker:** *.Models.Model.Bro ...
- Home Location:** *.Models.Model.Hor ...

2. The *Worker* then checks if the material flow object, to which the *Workplace* is assigned, has a **Scope** with the respective **Objects** in his dialog.

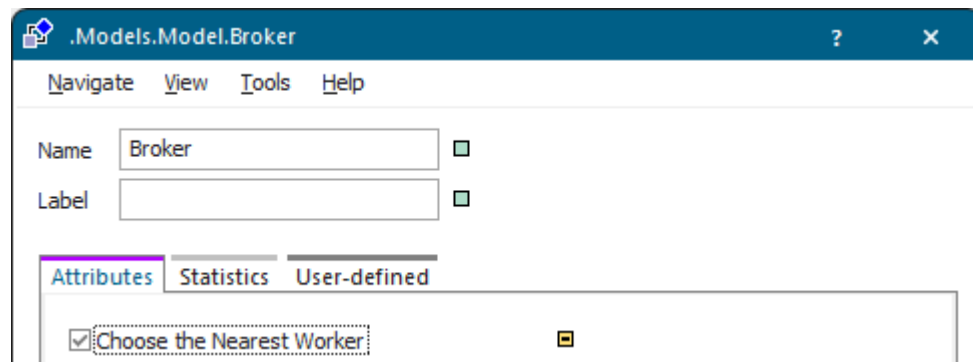
The screenshot shows the 'Scope' tab of the '.Resources.Worker:1' dialog. It contains the following fields and controls:

- Name:** Worker
- Label:** (empty)
- Number:** 1
- Stopped:** ☐
- Failed:** ☐
- Planned:** ☒ (selected)
- Objects:** (table with 3 rows)

	Objects
1	Station1
2	Station2
3	

If the current material flow object is part of the correct **Scope**, he stays on the current *Workplace* and finishes his current job there.

3. The respective *Broker* also checks if **Choose the Nearest Worker** is selected.



4. If the *Worker* has the highest **Priority**, a matching or no **Scope**, and the **shortest route** to the *Workplace*, the *Broker* mediates him to the respective *Workplace*.

In doing so, *Plant Simulation* computes a route to the *Workplace* for all eligible *Workers* while brokering them. The more *Workers* can be brokered, the more time it takes to compute the route to the *Workplace*. If you configured **additional services**, this might use up even more time. This can, consequently, slow down the simulation.

See also

[How the Broker Mediates Jobs to the Worker](#)

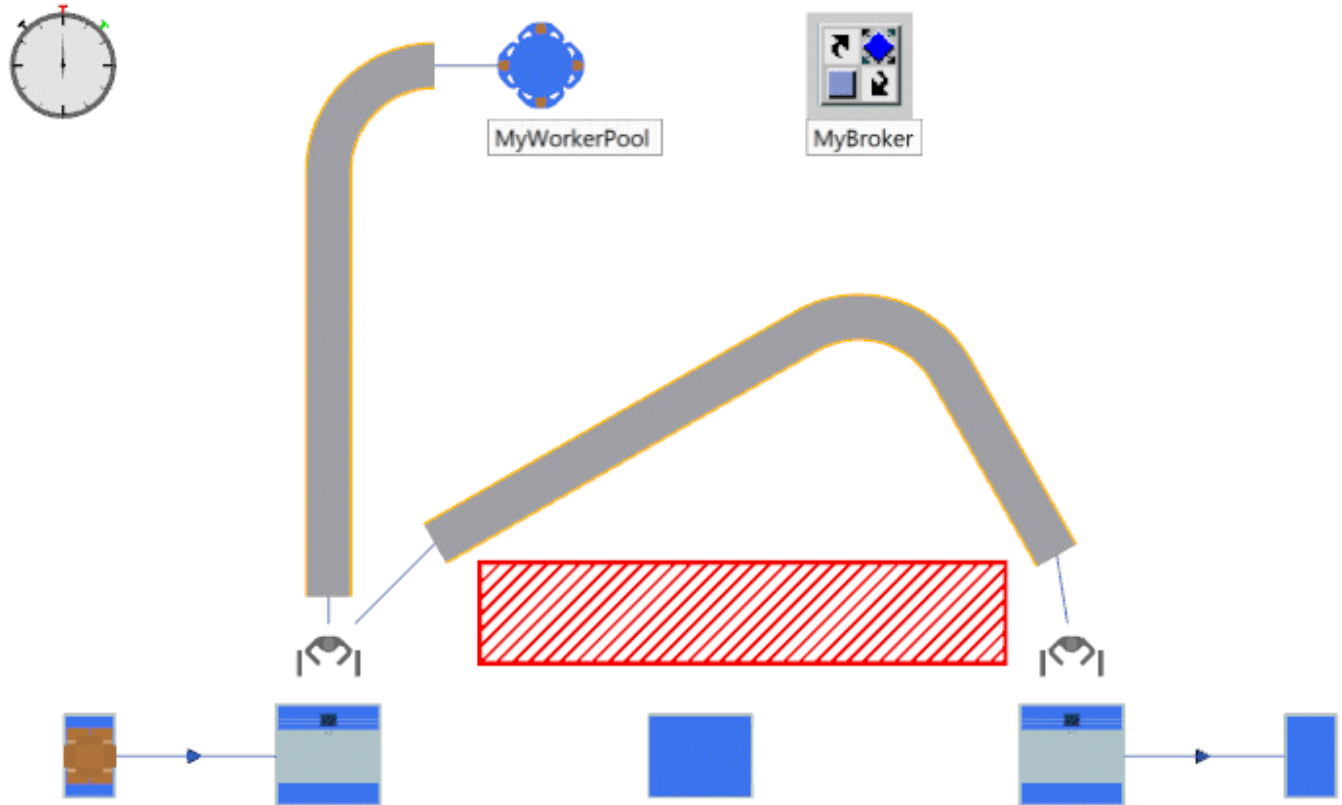
[Set Home Location and Scope for the Worker](#)

[Tab Scope](#)

[Choose the Nearest Worker \[check box\]](#)

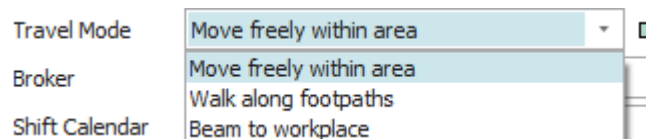
How the Worker Carries Parts

The *Worker* can carry parts from one station to another station. This only works if you attach a *Workplace* to the work station at which the *Worker* picks the part up. Also attach a *Workplace* to the target station at which he places the part. You also have to insert a *Broker* and a *WorkerPool*.



Remarks

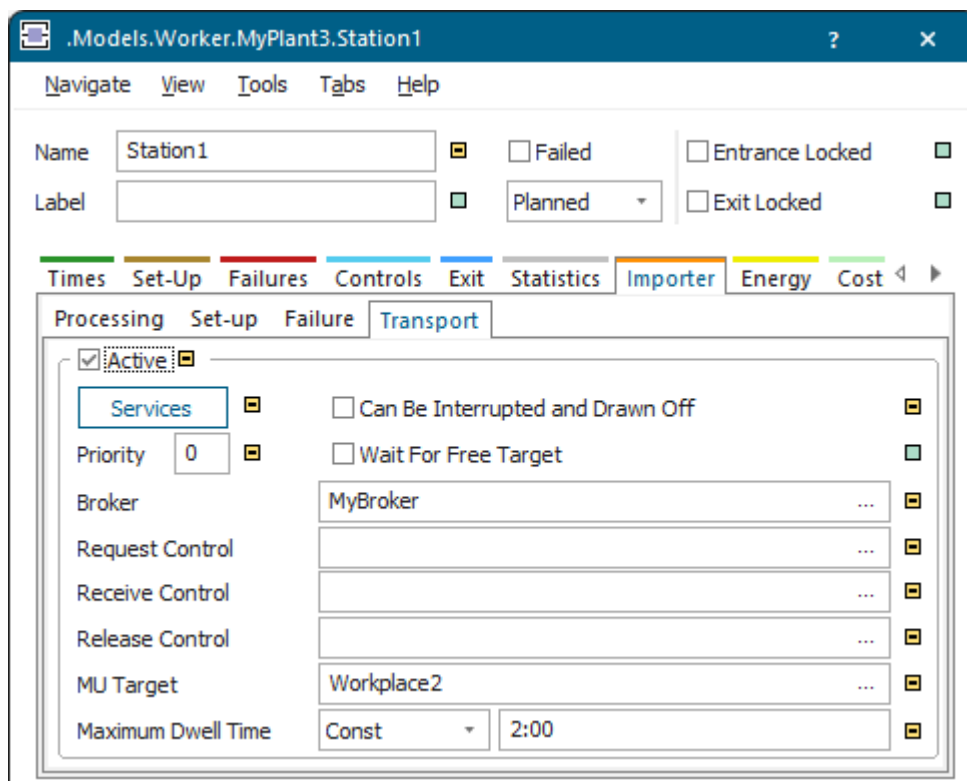
Select the **Travel Mode** in the **WorkerPool**:



- For the setting **Move freely within area** the *Worker* automatically finds his way to the destination of the part.
- For the setting **Walk along Footpaths** you have to connect the *Workplaces* with *Footpaths*. Also connect the *WorkerPool* and the *Workplace* at the station at which the *Worker* picks the part up, with a *Footpath*.

Activate the **transport importer** on the tab **Importer** > **sub-tab Transport** the dialog of the material flow object at which the *Worker* is to pick the part up. In the example below this is the station named *Station1*.

Then, specify the **Broker** who brokers the *Worker*. If the *part* itself does not have a destination, you also have to specify the **MU Target** to which he carries the *part*. The **MU Target** either is the *Workplace* attached to the *work station*, which processes the *part* next and to which the *Worker* carries the *part*, or the *work station* itself.



The *part* then requests a *Worker*. The *parts* wait at the station until the *Worker* arrives and picks them up:

- If the *Worker* can only carry a single *part*, if its capacity is 1 (**X-Dimension** = 1, **Y-Dimension** = 1, **Z-Dimension** = 1), he picks the *part* up and carries it to the destination.
- If the *Worker* can carry several *parts*, he will wait for parts until his capacity is reached. To prevent the *Worker* from waiting too long for additional *parts* to arrive, type in the **Maximum Dwell Time**. After it has passed, he moves to the target station, the **MU Target**, and places the *part*.

The *Worker* evaluates the **MU Targets** and moves to the closest one first. At the target station, he places all parts destined for this station. If he carries additional parts, he evaluates the next target and moves on to the closest one, until he has placed all *parts*.

If your plant works in shifts, the *Worker* walks back to his *WorkerPool* at the end of his shift. He then deposits the parts which he could not deliver into the **Parts Buffer** which you assigned to the *WorkerPool*. If you did not assign a *parts buffer* to the *WorkerPool*, *Plant Simulation* shows a warning and stops the simulation.

If you activated the *transport importer* in the *parts buffer*, one of the available *Workers* carries the parts in the *parts buffer* to the destination object after the next shift has started. If you did not activate the

transport importer, the *parts* in the *parts buffer* are moved on according to the **Exit Strategy** which you selected for the *parts buffer*.

See also

[Model a Worker Who Carries Parts Between Workplaces](#)

[Graphic Settings](#)

Dialog Box of the Worker

Dialog Box of the Worker

Double-click the icon of the *Worker* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

See also

[Stopped \[check box\] - Worker](#)

[Failed \[check box\] - material flow objects](#)

[Paused/Planned/Unplanned \[material flow objects\]](#)

[Number \[of the MU\]](#)

[States of the Worker](#)

Stopped [check box] - Worker

To stop the *Worker* on his way to or from the destination, select this check box. If the *Worker* stays in the *WorkerPool* or on a *Workplace*, you cannot stop him. To make the *Worker* continue on his way on the *Footpath* or if he is walking freely within the area, clear the check box.

☒ Stopped

SimTalk

Stopped [SimTalk] - Worker

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

The shared properties are described under the [Tab Attributes](#).

Priority [Worker]

Type the **Priority** into the text box with which the *Worker* carries out a work order. The **Priority** is a criterion for the urgency of a request.

Priority 

Remarks

The *Priority* is an integer value. The higher its value, the higher the urgency. *Plant Simulation* provides a *Worker* with *Priority* 10 before providing a *Worker* with *Priority* 1.

Workers with a higher **Priority** will be brokered earlier by the *Broker* for identical services than *Workers* with a lower priority. The **Priority** only applies to *Workers* who are registered with the same *Broker*.

If several qualified *Workers* with the same priority exist, who can provide the requested service, *Plant Simulation* first brokers the *Workers*, who already stay on a suitable *Workplace* at the station. After that *Workers* are brokered who already are at the station, but stay on the wrong *Workplace*. Then *Workers* are brokered who are not staying on a *Workplace*. Finally, those *Workers* are brokered who stay on a *Workplace* that is assigned to another station.

SimTalk

Priority [SimTalk] - Worker

Efficiency [text box]

Type the **Efficiency** in percent into the text box with which the *Worker* works. It determines how fast the *Worker* performs the work assigned to him.

Efficiency %

Remarks

Compare these examples:

- With an **Efficiency** of 100 percent the *Worker* consumes the exact processing time which you specified.
- With an **Efficiency** of 200 percent he consumes half of the processing time which you specified.
- With an **Efficiency** of 50 percent he consumes twice the processing time which you specified.

Note

If more than one *Worker* performs jobs at the station, the efficiency of the slowest *Worker* determines the processing time of the *MU*.

SimTalk

Efficiency [SimTalk] - Worker

See also

Workers to Create

Speed [Worker]

Type the **Speed** into the text box. This is the **Speed** with which the *Worker* walks on the *FootPath* or walks freely within the area to the designated *Workplace* at the work station.

Speed  m/s

SimTalk

Speed [SimTalk] - Worker

CurrentSpeed [SimTalk] - Worker

X-Dimension [Worker]

Type how many *parts* the *Worker* can carry in the **X-Dimension** into the text box.

X-Dimension 

Remarks

The **carrying capacity** of the *Worker* is **X-Dimension** times **Y-Dimension** times **Z-Dimension**. The greatest allowed value is one million.

If you decrease the carrying capacity of the *Worker*, make sure that no *parts* are located on the places that will be deleted by this action! Either delete these *parts* or move them to another place on the reduced carrying capacity area.

SimTalk

[XDim \[SimTalk\] - Worker](#)

See also

[Y-Dimension \[Worker\]](#)

[Z-Dimension \[Worker\]](#)

Y-Dimension [Worker]

Type how many *parts* the *Worker* can carry in the **Y-Dimension** into the text box.

Y-Dimension 

Remarks

The **carrying capacity** of the *Worker* is **X-Dimension** times **Y-Dimension** times **Z-Dimension**. The greatest allowed value is one million.

If you decrease the carrying capacity of the *Worker*, make sure that no *parts* are located on the places that will be deleted by this action! Either delete these *parts* or move them to another place on the reduced carrying capacity area.

SimTalk

[YDim \[SimTalk\] - Worker](#)

See also

[X-Dimension \[Worker\]](#)

[Z-Dimension \[Worker\]](#)

Z-Dimension [Worker]

Type how many *parts* the *Worker* can carry in the **Z-Dimension** into the text box.

Z-Dimension 

Remarks

The **carrying capacity** of the *Worker* is **X-Dimension** times **Y-Dimension** times **Z-Dimension**. The greatest allowed value is one million.

The **Z-Dimension** enables to stack parts one onto the other.

If you decrease the carrying capacity of the *Worker*, make sure that no *parts* are located on the places that will be deleted by this action! Either delete these *parts* or move them to another place on the reduced carrying capacity area.

SimTalk

[ZDim \[SimTalk\] - Worker](#)

See also

[X-Dimension \[Worker\]](#)

[Y-Dimension \[Worker\]](#)

Video on YouTube

<https://youtu.be/HiPziA8kxc0?si=tx1MOqgt0BsmOuxB&t=693>

Shift [Worker]

Type the name of the **Shift** into the text box during which the *Worker* works.

Shift 

Remarks

If you do not specify a specific shift on the [Tab Shift Times](#) of the [ShiftCalendar](#), the *Worker* works during all shifts defined in this *ShiftCalendar*.

SimTalk

Shift [SimTalk] - Worker

See also

Tab Shift Times

Services [Worker]

To open a list, into which you type the names of the services, which the *Worker* provides, click this button.



Remarks

Before you can type in data, click the **Inheritance** check box so that it looks like this .

- The *Worker* provides these services by default:

Services	Priority	Efficiency
Processing		
Set-up		
Failure		
Transport		
StandardService		

- Type the name of the service into the cell and click **Apply**. The *Worker* attempts to provide the services in the order in which you type them into the list. You might, for example, enter repair, drilling, milling, turning, carrying parts, etc.

Note

The name is not case-sensitive, just like the names of *attributes* and *methods* of the objects are not case-sensitive.

To save memory and improve access speed, all places which are using such a case-insensitive string are pointing to the same string in main memory. The visible and unexpected result is that the first occurrence of the string defines how the string is written in terms of upper- and lower-casing.

In *SimTalk* you can compare strings in an case-insensitive manner with the `~=` operator, compare [Relational Operators](#).

If you type in a **Priority** and an **Efficiency** for each individual service, *Plant Simulation* uses these values. By typing in a **Priority** and an **Efficiency**, you are better able to control the automatic assignment of the services, which the *Worker* provides for jobs to be done.

Services	Priority	Efficiency
StandardService		
repair	1	80.00

If you do not type in a **Priority** and an **Efficiency** for an individual service, *Plant Simulation* uses the values for the **Priority** and the **Efficiency** of the *Worker*.

Services	Priority	Efficiency
StandardService		
repair		

Plant Simulation inserts these **services** into the list of offered services of the *Broker*: To view them, select **View > Offered Services**.

You can use the commands on the [Context Menu of Embedded Lists](#) to edit the contents of the cells.

If you want the *Worker* to provide additional services, you can type in these services into the [Workers to Create](#) table.

	Worker	Amount	Shift	Speed	Efficiency	Home Location	Scope	Additional Services
1	*.Resources.John	2					-	repair
2	*.Resources.Nellie	1					-	-

SimTalk

[getServices \[SimTalk\] - Worker](#)

[setServices \[SimTalk\] - Worker](#)

[Services \[SimTalk\] - Worker](#)

See also

[Exported Services \[Worker\]](#)

[Offered Services \[Broker\]](#)

[Context Menu of Embedded Lists](#)

Video on YouTube

<https://youtu.be/HiPziA8kxc0?si=qYFCTtAc7V4GzXjQ&t=178>

Service [text box]

Shows the name of the brokered or of the assigned service for the *Worker* who carries a part. If the *Worker* is brokered to an *importer*, the text box shows the brokered service.

Service StandardService

Remarks

If the *Worker* carries parts to a **Destination**, he is no longer brokered. While the *Worker* carries parts, the assigned service is shown which corresponds to the brokered service during loading. The service remains assigned until the *Worker* has unloaded all parts.

While he is loading, the *Worker* is brokered. For this reason you can query this service during the **Loading Time** with the method *getExportedServices*.

See also

[Loading Time \[Workplace\]](#)

[getExportedServices \[SimTalk\]](#)

Broker [Worker]

Select the *Broker*, who assigns the services to the *Worker*.

Broker 

Remarks

Click the ellipsis button and select the *Broker*, who assigns the services to the *Worker*, in the dialog **Select Object**.

Or type in the path to and the name of the *Broker*. You might, for example, type in `.Models.Model.MyBroker`.

Or select the *Broker* in a *Frame*, drag it to the text box and drop it there.

Note

F2 opens the dialog of the object, whose name you typed into the text box.

SimTalk

BrokerPath [SimTalk] - Worker

See also

Broker [object]

Select Object [for controls]

Home Location

Select the **Home Location**. This is the *Workplace* to which the *Worker* walks when he has finished his current job.

Home Location 

Remarks

Click the ellipsis button and select the **Home Location** of the **Worker** in the dialog **Select Object**.

Or type in the path to and the name of the *Workplace*. You might, for example, type in `~.Models.Model.Workplace`.

Or select the **Home Location** in a *Frame*, drag it to the text box and drop it there.

Note

Plant Simulation transfers *Workers*, for whom you defined a **Home Location** and a **Shift** in the **Workers to Create** table of the *WorkerPool*, to the defined **Home Location** at the start of the **Shift**. They then start walking to the respective *Workplace* from there.

At the end of the shift *Plant Simulation* transfers all *Workers* of this shift directly to the *WorkerPool*, they do not walk back to the *WorkerPool*.

Note

Plant Simulation transfers *Workers*, for whom you defined a **Home Location** but no **Shift**, to the defined **Home Location** at the start of the simulation.

Note

Plant Simulation always resolves **relative paths** in relation to the *Frame* in which the *WorkerPool* is inserted. If *Plant Simulation* cannot resolve a path in the **Scope** array during the simulation, it shows an error message and asks if you would like to stop the simulation.

Note

F2 opens the dialog of the object, whose name you typed into the text box.

You can also enter the **Home Location** into the **Workers to Create** table. *Plant Simulation* then enters it into the text box **Home Location**.

 .Models.Model.WorkerPool - .Models.Model.WorkerPool.WorkersToCreateTable

1

	Worker	Amount	Shift	Speed	Efficiency	Home Location	Scope	Additional Services
1	*.Resources.Worker	1				Home1	Station1; Station2	-
2	*.Resources.Worker	1				Home11	Station11; Station21	

OK Cancel Apply

SimTalk

HomeLocation [SimTalk] - Worker

goTo [SimTalk]

goToHome [SimTalk]

goToPool [SimTalk]

teleportTo [SimTalk]

teleportToHome [SimTalk]

teleportToPool [SimTalk]

See also

Select Object [for controls]

Set Home Location and Scope for the Worker

Video on YouTube

https://youtu.be/gU1pqu7sWA8?si=dUIUQrzFfJ_XIAv7&t=38

Tab Failures

Define **failures** as described under the [Tab Failures](#).

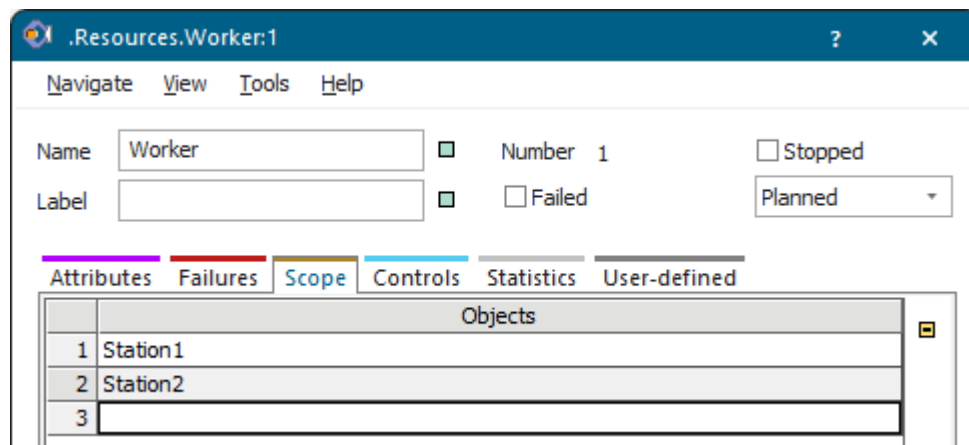
Tab Scope

Tab Scope

Assign any of the built-in [material flow objects](#), or a *Frame*, to which the *Worker* can be brokered to perform a job, on the tab **Scope**.

Remarks

To add an object to the scope of the *Worker* to the tab **Scope**, drag it from the *Frame* window over the icon of the *Worker* class and drop it there. You can also select and then drag-and-drop multiple objects at the same time!



Note

Plant Simulation always resolves [relative paths](#) in relation to the *Frame* in which the *WorkerPool* is inserted. If *Plant Simulation* cannot resolve a path in the **Scope** array during the simulation, it shows an error message and asks if you would like to stop the simulation.

To manipulate the contents of the list, use the commands on the [Context Menu of Embedded Lists](#).

You can also enter the **Scope** into the [Workers to Create](#) table.

 .Models.Model.WorkerPool - .Models.Model.WorkerPool.WorkersToCreateTable

1

	Worker	Amount	Shift	Speed	Efficiency	Home Location	Scope	Additional Services
1	*.Resources.Worker	1				Home1	Station1; Station2	-
2	*.Resources.Worker	1				Home11	Station11; Station21	

OK Cancel Apply

Plant Simulation then enters the respective stations into the cells in the **Objects** table.

See also

[Objects \[Scope\]](#)

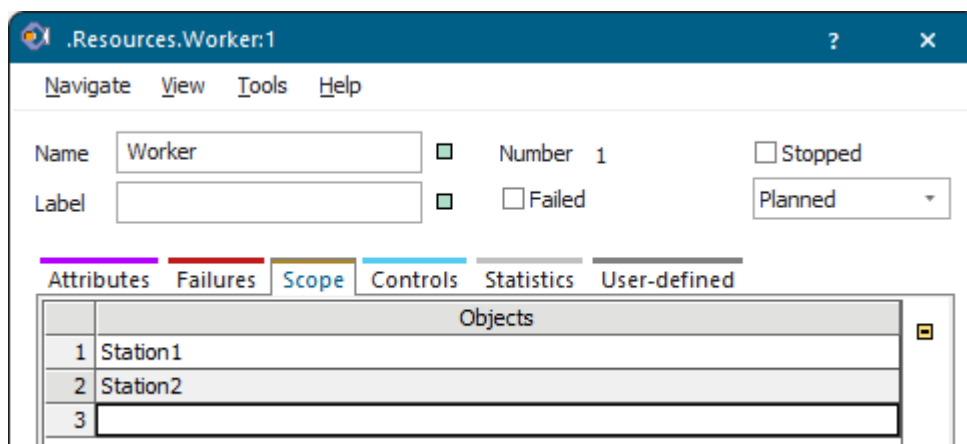
[How the Worker Decides Where to Work](#)

[Set Home Location and Scope for the Worker](#)

[Choose the Nearest Worker \[check box\]](#)

Objects [Scope]

Assign the objects, which form the **Scope**, i.e., any of the built-in material flow objects, or a *Frame*, to which the *Worker* can be mediated to do a job.



Remarks

Before you can type in data, click the **Inheritance** check box so that it looks like this .

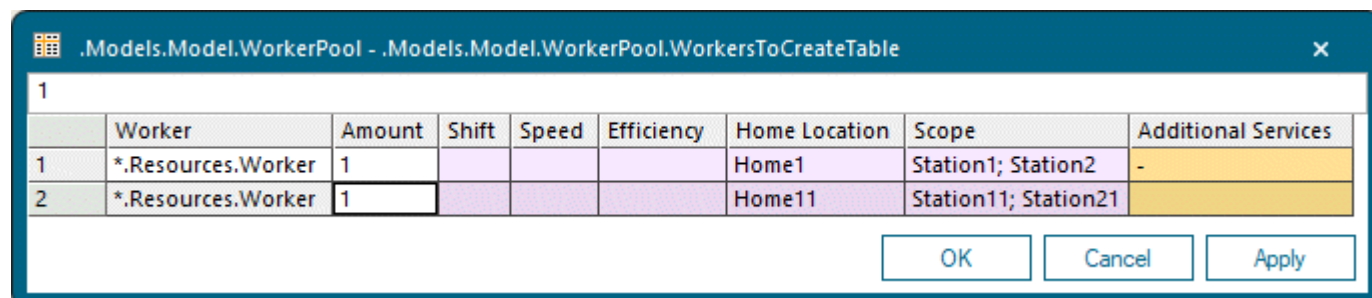
- Type in the path to and the name of the material flow object in a cell. Or
- Drag the material flow object to the icon of the *Worker* and drop it there.
- Select several material flow objects and drag them to the icon of the *Worker* and drop them there.

Note

Plant Simulation always resolves **relative paths** in relation to the *Frame* in which the *WorkerPool* is inserted. If *Plant Simulation* cannot resolve a path in the **Scope** array during the simulation, it shows an error message and asks if you would like to stop the simulation.

To allow the *Worker* to be mediated to a *Frame*, including all of the inserted objects and *sub-Frames*, specify that *Frame*.

You can also enter the **Scope** into the **Workers to Create** table.



Plant Simulation then enters the respective stations into the cells in the **Objects** table.

SimTalk

[Scope \[SimTalk\] - Worker](#)

[setWorkersToCreateTable \[SimTalk\]](#)

See also

[How the Worker Decides Where to Work](#)

[Set Home Location and Scope for the Worker](#)

[Choose the Nearest Worker \[check box\]](#)

Tab Controls

Tab Controls [Worker]

Click the ellipsis button and select a *Method* in the dialog **Select Object**. Type in the source code of the respective control.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

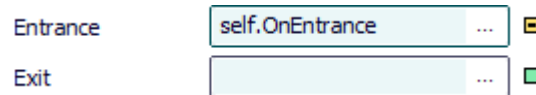
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

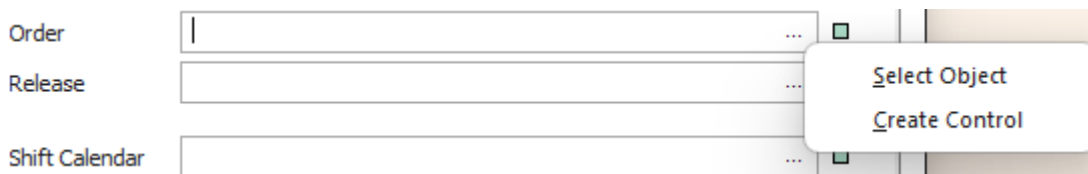
See also

[Order Control \[Worker\]](#)

[Release Control \[Worker\]](#)

Order Control [Worker]

Modifies the built-in behavior of the object. The **Order Control** specifies how the *Worker* handles the order. The *Worker* calls the **Order Control** whenever he is assigned to an *importer*.



Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



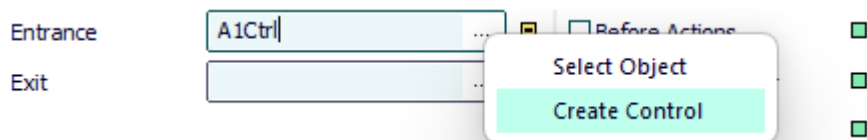
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

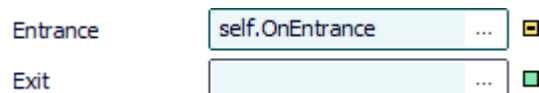
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.

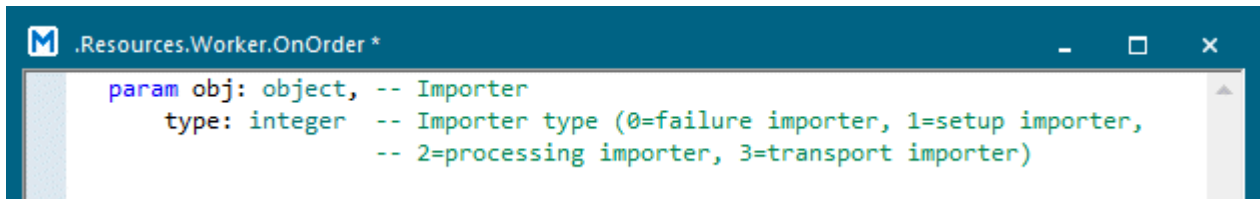


Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

The **standard order control** as a *user-defined attribute* looks like this:



Parameters

The **Order Control** has two parameters, which define the *importer*:

- The parameter *Importer* of data type *object* designates the *importer*.
- The parameter *Type* of data type *integer* designates its type: 0 designates the *failure/remove failure-importer*, 1 the *set-up-importer*, 2 the *processing-importer*, and 3 the *transport-importer*.

SimTalk

OrderCtrl [SimTalk] - Worker

See also

[Select Object \[for controls\]](#)

Release Control [Worker]

Modifies the built-in behavior of the object. The object calls the **Release Control** as soon as an *importer* releases the *Exporter/Worker*.

Remarks

Plant Simulation executes the **Release Control** as soon as an *importer* releases the *Exporter/Worker*. When the **Release Control** is called, the *Exporter/Worker* has already left its *importer*. You yourself have to make sure that the *Exporter/Worker* is assigned new *importers*, for example with the method [findNewImporter](#). In this case the *Exporter/Worker* will not automatically register as being available with its *Broker*.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.

Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.

- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.

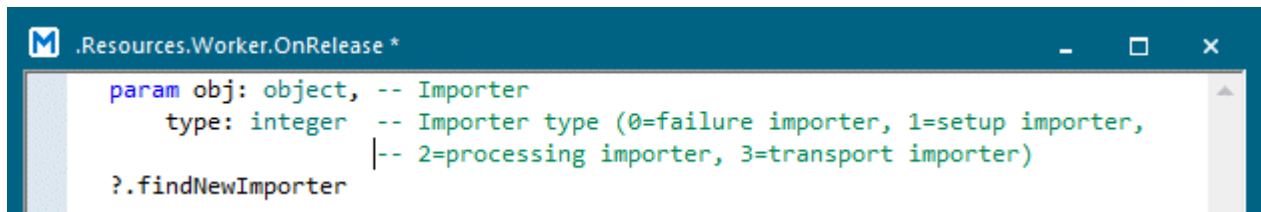
Entrance	self.OnEntrance	...	📄
Exit		...	📄

Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

The **standard release control** as a *user-defined attribute* looks like this:



```

param obj: object, -- Importer
      type: integer -- Importer type (0=failure importer, 1=setup importer,
                                |-- 2=processing importer, 3=transport importer)
?.findNewImporter
  
```

Parameters

The **Release Control** has two parameters, which define the *importer*:

- The parameter *Importer* of data type *object* designates the *importer* proper.
- The parameter *Type* of data type *integer* designates its type:
0 designates the *failure/remove failure-importer*, 1 the *set-up-importer*, 2 the *processing-importer*, and 3 the *transport-importer*.

Note

Plant Simulation calls the **Release Control**, which you typed into the *Worker* for the *transport-importer*, when the *Worker* starts to carry parts away. This is then the case if either the **Maximum Dwell Time [transport importer]** has passed or if the *Worker* cannot pick up any more parts because his **X-Dimension [Worker]**, **Y-Dimension [Worker]**, and **Z-Dimension [Worker]** is used up.

SimTalk

ReleaseCtrl [SimTalk] - Worker

See also

[Select Object \[for controls\]](#)

Tab Statistics

Tab Statistics [Worker]

Plant Simulation shows the most important statistical data on the tab **Statistics**.

Remarks

The tab **Statistics** shows the values for the *Services* and for the *Worker*. For each of these blocks, adding up the individual values results in 100 percent.

Attributes	Failures	Scope	Controls	Statistics	User-defined
☒ Exporter Statistics ☐					
Operational + Failed			Exporter		
Working	0.00%		Operational	100.00%	
Setting-up	0.00%		Paused	0.00%	
Repairing	0.00%		Unplanned	0.00%	
Transporting	28.05%		Failed	0.00%	
En-route to Job	13.39%		Worker		
Waiting	58.56%		Traveled Distance	212.1 m	
Failed	0.00%				

The *Worker* collects *failed* times of the *services* and of the *Worker*. The *Worker* only collects *Failed* times that accrue during the processing time, i.e., times that are located outside of the *paused* and of the *unplanned* time.

Waiting times only accumulate during the times if the *Worker* is available, i.e., if these times are located outside of the *paused*, the *unplanned*, and of the *failed* times.

Item English	Description	Read-only attribute	Item German
Services Setting-up	Shows the portion of the set-up time of the services of the overall statistics time of the <i>Worker</i> . The overall statistics time is the statistics collection period without the unplanned time and the paused time .	StatServicesSetupPortion [SimTalk] - Worker	Dienste rüstend
Services Processing	Shows the portion of the working time of the services of the overall statistics time of the <i>Worker</i> . The overall statistics time is	StatServicesWorkingPortion [SimTalk] - Worker	Dienste arbeitend

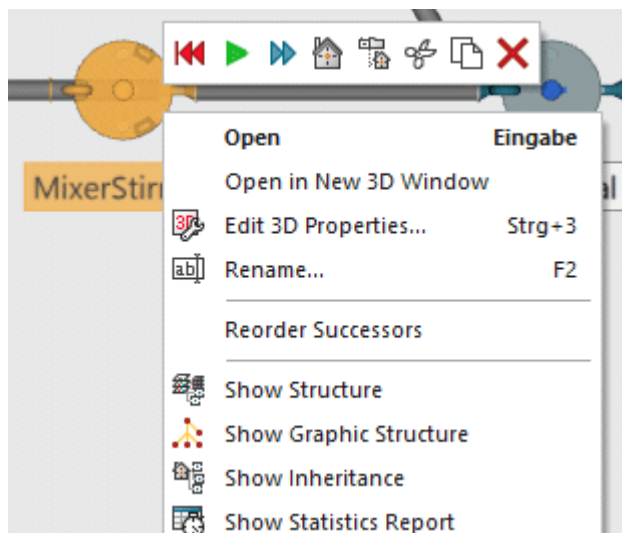
Item English	Description	Read-only attribute	Item German
	the statistics collection period without the unplanned time and the paused time .		
Services Repairing	Shows the portion of the repairing time of the services of the overall statistics time of the <i>Worker</i> . The overall statistics time is the statistics collection period without the unplanned time and the paused time .	StatServicesRepairingPortion [SimTalk] - Worker	Dienste reparierend
Services Transporting	Shows the portion of the transporting time from <i>Workplace</i> to <i>Workplace</i> of the services of the overall statistics time of the <i>Worker</i> . The overall statistics time is the statistics collection period without the unplanned time and the paused time .	StatServicesTransportingPortion [SimTalk]	Dienste tragend
Services En-route to Job	Shows the portion of the time of the services of the overall statistics of the <i>Worker</i> spent for getting to the <i>Workplace</i> or back to the <i>WorkerPool</i> to get the next job order. The latter only applies if you select Get job orders at home only . The overall statistics time is the statistics collection period without the unplanned time and the paused time .	StatServicesEnRouteToJobPortion [SimTalk]	Dienste unterwegs zu Auftrag
Services Waiting	Shows the portion of the waiting time of the services for an <i>importer</i> plus the time the <i>services</i> spent waiting for a <i>MU</i> at the <i>importer</i> of the overall statistics time of the <i>Worker</i> . The value of StatServicesEnRouteIdlePortion is part of the value of StatServicesWaitingImpPortion . The overall statistics time is the statistics collection period without the unplanned time and the paused time .	StatServicesWaitingPortion [SimTalk] - Worker StatServicesWaitingImpPortion [SimTalk] - Worker StatServicesWaitingMUPortion [SimTalk] - Worker StatServicesEnRouteIdlePortion [SimTalk]	Dienste wartend
Services Failed	Shows the portion of the failed time of the services of the overall statistics time of the <i>Worker</i> . The overall statistics time is the statistics collection period without the unplanned time and the paused time .	StatServicesFailedPortion [SimTalk] - Worker	Dienste gestört
Exporter Operational	Shows the portion of the statistics collection period during which the <i>Worker</i> was operational.	StatExporterOperationalPortion [SimTalk]	Exporter einsatzbereit

Item English	Description	Read-only attribute	Item German
Exporter Paused	Shows the portion of the statistics collection period during which the <i>Worker</i> was paused.	StatExporterPausedPortion	Exporter pausiert
Exporter Unplanned	Shows the portion of the statistics collection period during which the <i>Worker</i> was unplanned.	StatExporterUnplanned Portion [SimTalk]	Exporter ungeplant
Exporter Failed	Shows the portion of the statistics collection period during which the <i>Worker</i> was failed.	StatExporterFailedPortion [SimTalk]	Exporter gestört
Worker Traveled Distance	Shows the distance which the instance of the <i>Worker</i> traveled from the <i>WorkerPool</i> to the <i>Workplaces</i> attached to the stations and between the stations in meters.	StatTraveledDistance [SimTalk] - Worker	Werker zurückgelegte Wegstrecke

To collect statistics data, select the check box [Exporter Statistics](#).

To view [Exporter and Worker Statistics](#) in the [Statistics Report](#), select **View > Show Statistics Report** in the dialog of the object. You can also click the right mouse button in the *Frame* and select [Show Statistics Report](#) or you can press **F6**.

You can also click the object in the *Frame* with the right mouse button and select [Show Statistics Report](#) on the context menu.



The individual states look like this:

Non-operational			Operational							
Unplanned	Paused	Failed	Working				Waiting			
			Setting-up	Processing	Repairing	Transporting	Waiting for MU	Waiting for importer		En-route to job
								Not en-route	En-route while idle	

Note
The *Worker* stays at the station, to which he is brokered, while this station is **failed**, **paused**, or **stopped**. Statistics does not count this time as **waiting**.

See also

[Tab Statistics \[WorkerPool\]](#)

Exporter Statistics [Worker]

To collect statistics data of the *Worker*, select this check box. To deactivate statistics collection, clear the check box.

☒ Exporter Statistics ☐

SimTalk

[ExpStatOn \[SimTalk\]](#)

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

View Menu [Worker]

The **View Menu** provides commands to access its functions.

Refresh [on View menu]	Importers [Worker]
Show Statistics Report [on View menu]	Exported Services [Worker]
Show Attributes and Methods [on View menu]	Route to Destination [Worker]
Contents [Worker]	Associated Shift Calendar
Position [Worker]	

See also

[View Menu \[general description\]](#)

Contents [Worker]

Opens a table, which shows all *MUs* which the *Worker* carries.

Remarks

For a *Worker* with the capacity 1 the table has one column. It has as many columns as the **capacity** (**X-Dimension**, the **Y-Dimension**, and the **Z-Dimension**) you typed in.

1	
1	2
2	2

1	
1	*.MUs.Entity:439725

SimTalk

Cont [SimTalk] - MUs

contentsList [SimTalk] - material flow objects

deleteMovables [SimTalk] - MUs

mu [SimTalk] - material flow objects

muPart [SimTalk]

NumMU [SimTalk] - MUs

NumMUParts [SimTalk] - MUs

pe(Y,Y) / [X,Y] [SimTalk] - MUs

Position [Worker]

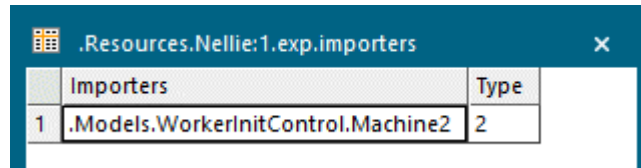
Opens a table, which shows the object on which the *Worker* is located at the moment.



Position	
1	*.Models.WorkerInitControl.Workplace2

Importers [Worker]

Opens a table, which shows all *importers* for which the *Worker* provides services.



	Importers	Type
1	.Models.WorkerInitControl.Machine2	2

Remarks

The table shows the **Importers** in column 1, and their **Type** in column 2: The number 0 designates the *failure-importer*, 1 the *set-up-importer*, 2 the *processing-importer*, and 3 the *transport-importer*.

SimTalk

[findNewImporter \[SimTalk\]](#)

[getImporters \[SimTalk\]](#)

Exported Services [Worker]

Opens a table, which shows the names of all services which the *Worker* provides at this time.

Double-click the name of a service to open a subtable, which shows the name of the **Importer**, the **Type** of the *importer* and the **Amount** of services provided.

The **Type** 0 designates the *failure-importer*, 1 the *set-up-importer*, 2 the *processing-importer*, and 3 the *transport-importer*.

.Resources.John:2.exp.exportedServices			
	Services		
1	StandardService		

.Resources.John:2.exp.exportedServices[1,1]			
	Importers	Type	Amount
1	.Models.WorkerInitControl.Machine1	2	1

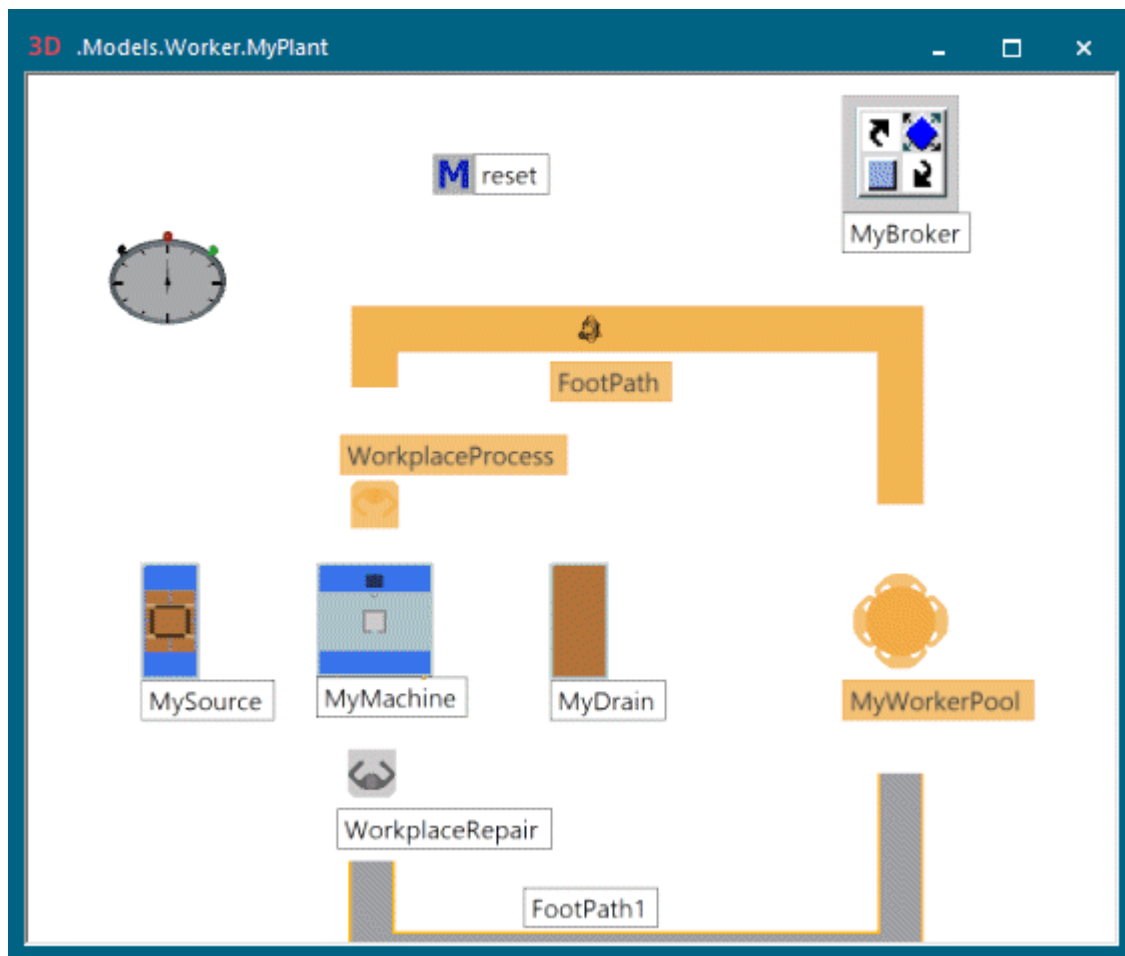
SimTalk

getExportedServices [SimTalk]

Services [SimTalk] - Worker

Route to Destination [Worker]

Selects all objects along the route to the destination object of the selected instance of the *Worker*.



See also

[getRouteLength \[SimTalk\] - Worker](#)

[getRouteCoordinates \[SimTalk\]](#)

[getRouteLength \[SimTalk\] - Workplace](#)

Tools Menu

The **Tools Menu** provides these menu commands:

Edit Controls

Edit Observers

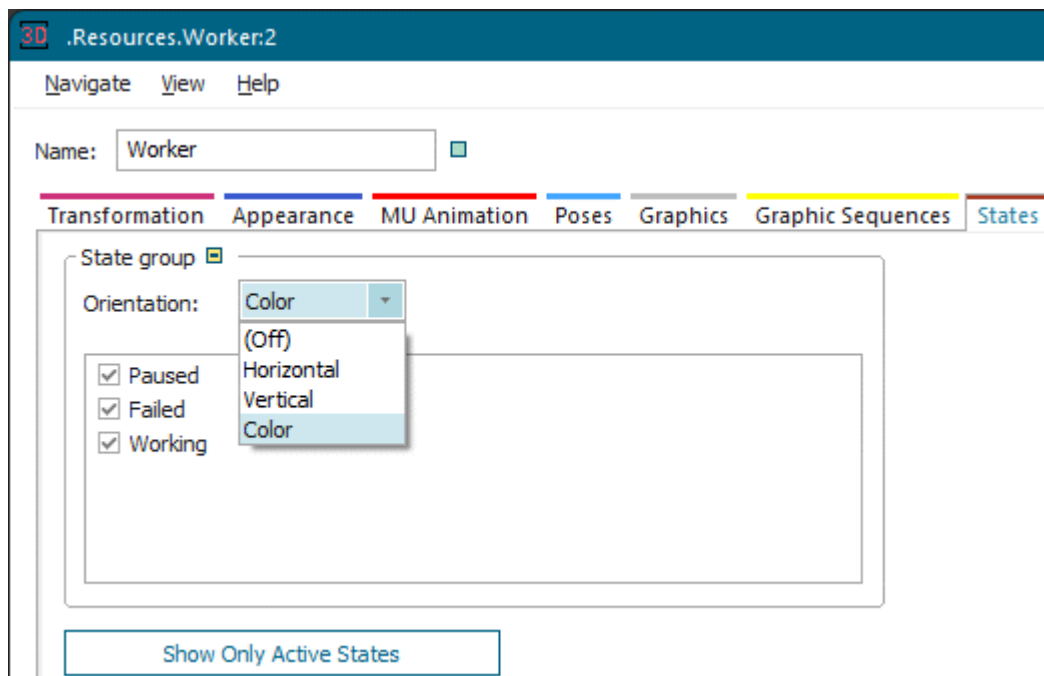
In addition, the *Worker* provides the **Available control** and the **Not-available control**.

Help Menu

The commands are described under the [Help Menu](#).

States of the Worker

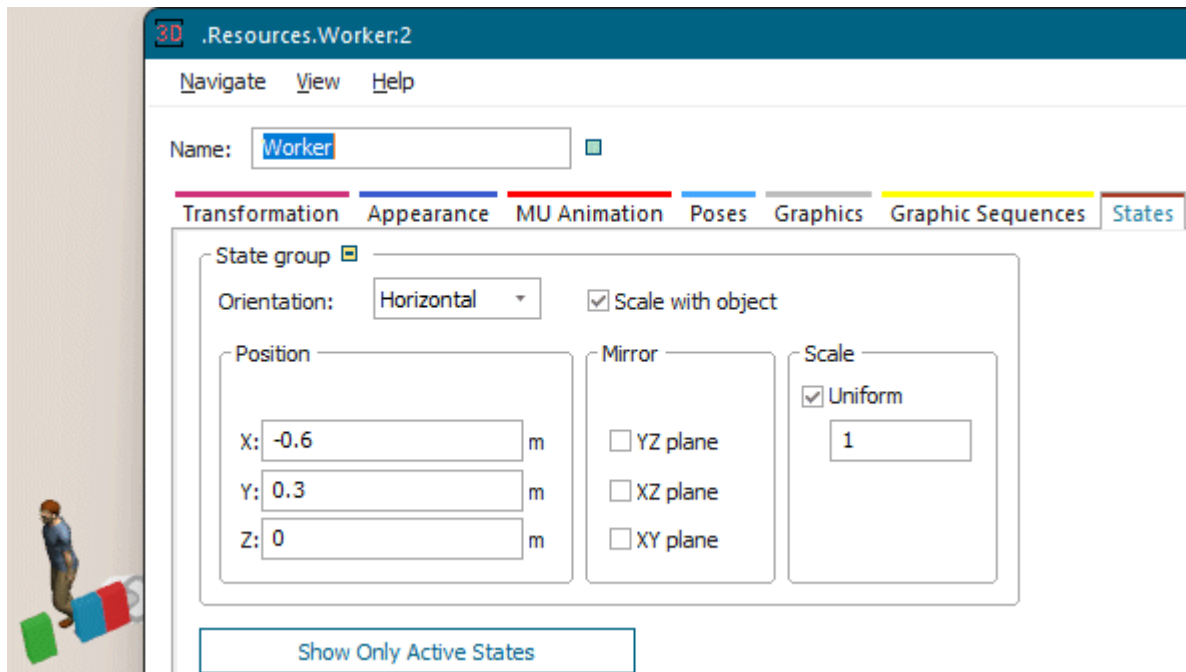
During your simulation runs *Plant Simulation* can show the state of the *Worker*. Select a setting on the **Tab States > States Orientation**.



- The setting **Horizontal** shows the state as a **horizontally arranged cube** along the bottom of his picture.



For the setting **Horizontal** you can show all states which the *Worker* can take.



- The setting **Color** colors the entire picture of the *Worker*.



See also

[States > State Graphic \[defined\]](#)

[States of the Material Flow Objects](#)

[States of the Energy Features](#)

[States of the Frame](#)

[States of the Transporter](#)

[States of the Method](#)

[States of the MQTT Interface](#)

Methods of the Worker

_Methods of the Worker

The *Worker* provides:

- The *methods* listed in the table of contents to the left.

Note

If you do not use a [FootPath](#), *Plant Simulation* beams/teleports the *Worker* to the *Workplace*.

- The [_Methods of the Exporter](#) as the *Worker* is an [Exporter \[object\]](#) with the capacity 1.
- The [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Figure 1: Methods of the Worker (Station)

Name	Value	Inherited	Watchable	Signature
contentsList	[]			(([Contents:table]) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				((IconName:string, Width:integer, Height:integer]) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method Identifier and data type of the parameter you enter Data type of the return value

Figure 2: Methods of the Worker (Station)

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			((Index:integer:=1]) -> object
MUPart	VOID			((Index:integer:=1]) -> object

Name of the method Identifier and data type of the parameter you enter Default value Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).